

Supplementary Material 2: Performance Evaluation Metrics

Description: This Supplementary material provides the mathematical definitions for the performance metrics employed to evaluate the proposed consistent SLAM approach.

1. Mean of the position estimation error's norm:

$$\bar{e}_{\mathbf{p},exp} = \frac{1}{N} \sum_{k=1}^N \|\mathbf{p}_{k,exp} - \hat{\mathbf{p}}_{k,exp}\|_2, \quad (1)$$

where $\mathbf{p}_{k,exp} = (x_{k,exp}, y_{k,exp})$ represents the mobile's robot position ground truth for sample k and experiment run exp . Similarly, $\hat{\mathbf{p}}_{k,exp} = (\hat{x}_{k,exp}, \hat{y}_{k,exp})$ is the corresponding estimated position, N is the total number of trajectory samples in experiment exp .

2. Mean of the angular estimation error's absolute value:

$$\bar{e}_{\theta,exp} = \frac{1}{N} \sum_{k=1}^N |\theta_{k,exp} - \hat{\theta}_{k,exp}|, \quad (2)$$

where $\theta_{k,exp}$ represents the mobile robot's angular orientation ground truth and $\hat{\theta}_{k,exp}$ is the estimated robot heading.

3. Mean estimation errors of the positional displacement magnitude:

$$\bar{e}_{\Delta\mathbf{p},exp} = \frac{1}{N-1} \sum_{k=1}^{N-1} \|\mathbf{p}_{k+1,exp} - \mathbf{p}_{k,exp}\| - \|\hat{\mathbf{p}}_{k+1,exp} - \hat{\mathbf{p}}_{k,exp}\| \quad (3)$$

4. Mean estimation errors of the angular displacement magnitude:

$$\bar{e}_{\Delta\theta,exp} = \frac{1}{N-1} \sum_{k=1}^{N-1} |\theta_{k+1,exp} - \theta_{k,exp}| - |\hat{\theta}_{k+1,exp} - \hat{\theta}_{k,exp}| \quad (4)$$

5. Root mean square estimation error for variable v :

$$e_{RMS_{v,exp}} = \sqrt{\frac{1}{N} \sum_{k=1}^N \|v_{k,exp} - \hat{v}_{k,exp}\|^2}, \quad (5)$$

where $v_{k,exp}$ is the ground truth value of the variable v at sampling instant k and experiment run exp and $\hat{v}_{k,exp}$ is the estimated value of the variable v at the corresponding instant. The RMS values of the estimation errors are computed for the position $\mathbf{p}_{k,exp} = (x_{k,exp}, y_{k,exp})$ and heading $\theta_{k,exp}$ variables, as well as for the magnitudes of the estimated translational and angular displacements $\Delta\mathbf{p}_{k,exp} = \|\mathbf{p}_{k+1,exp} - \mathbf{p}_{k,exp}\|$ and $\Delta\theta_{k,exp} = |\theta_{k+1,exp} - \theta_{k,exp}|$, i.e. $v_{k,exp} = \mathbf{p}_{k,exp}, \theta_{k,exp}, \Delta\mathbf{p}_{k,exp}$ or $\Delta\theta_{k,exp}$.

6. Positional error over path length percentage:

$$e_{p\%exp} = 100 \cdot \frac{\bar{e}_{\mathbf{p},exp}}{d_{exp}} [\%], \quad (6)$$

where d_{exp} is the total distance along the trajectory executed in experiment exp .

7. End-point estimation error:

$$e_{end,exp} = \|\mathbf{p}_{N,exp} - \hat{\mathbf{p}}_{N,exp}\|_2 \quad (7)$$

measures the distance between the end-point $\mathbf{p}_{N,exp}$ of the executed trajectory and the estimated end-point $\hat{\mathbf{p}}_{N,exp}$. This metric is useful to reveal possible divergence of SLAM algorithms with respect to the ground truth if the algorithm repeatedly fails to solve localization correctly.

8. *Map precision (positive predictive value):*

$$Precision = \frac{TP}{TP + FP}, \quad (8)$$

where TP is the total amount of true positives and FP is the total amount of false positives counted by comparing the estimated map to the ground truth map. Specifically,

$$\begin{aligned} TP &= \sum_{i=1}^{N_i} \sum_{j=1}^{N_j} \mathbb{1}((i, j) \in \mathcal{TP}), \\ FP &= \sum_{i=1}^{N_i} \sum_{j=1}^{N_j} \mathbb{1}((i, j) \in \mathcal{FP}), \end{aligned}$$

where the sets $\mathcal{TP}$ and $\mathcal{FP}$ of true positive and false positive cells in the map are respectively given by:

$$\begin{aligned} \mathcal{TP} &= \{(i, j) | P(\hat{\mathbf{m}}_{i,j,t}) \geq P(\mathbf{m}_{i,j,t}) \geq p_{occ}, \hat{\mathbf{m}}_{i,j,t} \in \hat{\mathcal{M}}_t, \mathbf{m}_{i,j,t} \in \mathcal{M}_t\}, \\ \mathcal{FP} &= \{(i, j) | P(\hat{\mathbf{m}}_{i,j,t}) \geq p_{occ} > P(\mathbf{m}_{i,j,t}), \hat{\mathbf{m}}_{i,j,t} \in \hat{\mathcal{M}}_t, \mathbf{m}_{i,j,t} \in \mathcal{M}_t\} \end{aligned}$$

for some predefined occupancy probability threshold that p_{occ} that the cell is occupied (typically p_{occ} is a value larger than 0.5). The precision or positive predictive value can be interpreted as indicating the accuracy in not mistakenly estimating that a cell is occupied when it is not since the precision value decreases as the number of false positives (false detections) increases.

9. *Map sensitivity (true positive rate or recall):*

$$Sensitivity = \frac{TP}{TP + FN}, \quad (9)$$

where TP is the total amount of true positives already introduced and FN is the total count of false negatives. Specifically,

$$FN = \sum_{i=1}^{N_i} \sum_{j=1}^{N_j} \mathbb{1}((i, j) \in \mathcal{FN}),$$

where the set $\mathcal{FN}$ of false negative map cells is given by:

$$\mathcal{FN} = \{(i, j) | P(\mathbf{m}_{i,j,t}) \geq p_{occ} > P(\hat{\mathbf{m}}_{i,j,t}), \hat{\mathbf{m}}_{i,j,t} \in \hat{\mathcal{M}}_t, \mathbf{m}_{i,j,t} \in \mathcal{M}_t\}$$

The sensitivity or recall can be interpreted as indicating the accuracy in detecting objects when the cell is occupied. The sensitivity decreases as the number of misdetections (false negatives) increases.

10. *Map accuracy:*

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} = \frac{TP + TN}{N_i \cdot N_j}, \quad (10)$$

corresponds to the ratio of total correctly estimated occupancy states over the total number $N_i \cdot N_j$ of map cells, i.e. an occupied grid cell that is correctly identified as occupied is a true positive, while a free (empty) grid cell that is correctly identified as free is a true negative. The total count of true negative cells is:

$$TN = \sum_{i=1}^{N_i} \sum_{j=1}^{N_j} \mathbb{1}((i, j) \in \mathcal{TN}),$$

where the set of true negative map cells is given by:

$$\mathcal{TN} = \{(i, j) | p_{occ} \geq P(\mathbf{m}_{i,j,t}) \geq P(\hat{\mathbf{m}}_{i,j,t}), \hat{\mathbf{m}}_{i,j,t} \in \hat{\mathcal{M}}_t, \mathbf{m}_{i,j,t} \in \mathcal{M}_t\}$$

The accuracy is sometimes also referred to as *simple matching coefficient* (SMC) or *Rand similarity coefficient*. Accuracy is a better metric for comparing the similarity between the ground truth map and the estimated map since it counts the cells that are mutually free or mutually occupied in the estimated map and the ground truth map, unlike the Jaccard index that only counts the true positives. It is to be noted that some authors employ other raster map similarity metrics, like mean absolute error (MAE) or RMS errors. The MAE is equal to the total amount of FP and FN over the total, which in the case of a binary occupancy map, is equivalent to computing $1 - \text{Accuracy}$. Thus, in this work the quality of the estimated map is reported in terms of accuracy (10) only and not MAE or the similar map RMS estimation errors.

If $\bar{v}_{exp}$ represents the value of any of the previous metrics obtained with data from experiment exp , then an estimator $\bar{v}$ for the value of the metric is computed by averaging the realizations and the estimator's corresponding confidence interval $[\bar{v} - C_v, \bar{v} + C_v]$ with confidence level 95% is provided considering the standard deviation $\sigma_{\bar{v}}$ of the realizations of $\bar{v}_{exp}$, $exp = 1, 2, \dots, N_e$, where N_e is the number of experiment realizations and

$$\begin{aligned} \bar{v} &= \frac{1}{N_e} \sum_{exp=1}^{N_e} \bar{v}_{exp}, \\ \sigma_{\bar{v}} &= \sqrt{\frac{1}{N_e} \sum_{exp=1}^{N_e} (\bar{v}_{exp} - \bar{v})^2}, \\ C_v &= \frac{1.96\sigma_v}{\sqrt{N_e}}. \end{aligned}$$

It is to be noted that the performance of SLAM methods is in general reported in terms of translation and angular displacement estimation errors in equations (3) and (4). However, the mean pose errors in coordinates of the world reference frame, expressed in equations(1) and (2), are also important to adequately compare SLAM algorithms employed in practical field robotics applications.